Runze Gu

Ph.D. Student in Civil Engineering | The University of Texas at Austin
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Education

The University of Texas at Austin
Ph.D. student in Civil Engineering; Graduate Research Assistant.
Advisor: Prof. Salvatore Salamone, Smart Structures Research Laboratory.

University of California, Berkeley - January-May 2024
GLOBE Engineering Visiting Student Program. GPA: 4.00/4.00.

Tongji University - September 2020-July 2024
Bachelor's degree in Civil Engineering. Overall grade: 89.89/100.
Outstanding Student Scholarship, three awards.

Research experience

Graduate Research Assistant - UT Austin
Smart Structures Research Laboratory, Department of Civil, Architectural and Environmental Engineering. Advisor: Prof. Salvatore Salamone.

Sustainable cement materials - UC Berkeley, March-May 2024
Advisor: Prof. Shaofan Li. Thermal reactivation of cement waste; heating-chamber development, sample preparation, and organization of materials for characterization.

Computer vision for seismic resilience assessment - Tongji University, January-July 2024
Advisor: Prof. Ying Zhou. Mask R-CNN-based building-damage assessment from satellite imagery for regional seismic resilience analysis.

Weld-joint detection for robotic welding - North Carolina State University, July-August 2023
Advisors: Prof. Kevin Han and Dr. Doyun Lee. Image annotation and U-Net-based weld-joint segmentation.

Nighttime vehicle detection - Tongji University, March 2022-December 2023
Advisor: Prof. Chao Liu. CycleGAN-based image translation and YOLOv5-based detection for low-light traffic imagery.

Professional experience and service

College of Civil Engineering, Tongji University - September 2024-May 2025
Core Officer, Department of International Activities and Student Affairs.

Vibration assessment and isolation design - September-December 2024
Structural modeling and vibration assessment with ETABS for a transit-oriented development project.

China Coal Xi'an Design Engineering Co., Ltd. - Summers 2022 and 2023
Construction supervision and inspection, including site documentation for metro infrastructure.

Honors

- Cockrell School of Engineering fellowship recipient, UT Austin.
- First-Class Outstanding Student Scholarship, Tongji University, 2022 and 2023.
- Second-Class Outstanding Student Scholarship, Tongji University, 2021.
- First Prize, "Sinopharm Cup" Chemistry Competition.
- Scholarship for Social Work, 2021.

Technical skills

- **Programming:** Python, MATLAB, Scheme.
- **Methods:** computer vision, machine learning, and structural analysis.
- **Engineering software:** ETABS, AutoCAD, and Revit.